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Fiscal decentralisation and economic growth in Vietnam

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By making use of a recently released panel dataset that covers 61 provinces of Vietnam, this paper examines the link between fiscal decentralisation and economic growth. Significant fiscal decentralisation has taken place in Vietnam from the mid-1990s. This paper explicitly considers the effect of fiscal decentralisation on economic growth (1) under the 1996 State Budget Law, which covered the period 1996 to 2001, and (2) under the 2002 State Budget Law, which covered the period 2002 to 2007. Our empirical analysis reveals that economic growth in Vietnam is positively associated with revenue decentralisation but negatively associated with expenditure decentralisation.

Keywords: fiscal decentralisation; subnational governments; economic growth

JEL classifications: C33, H77, O47

1. Introduction

Fiscal decentralisation refers to the transfer of authority for public spending and revenue collection from national to subnational governments. Owing to its practical importance, fiscal decentralisation remains an issue of intense debate among policy makers and researchers.¹ Fiscal decentralisation can lead to better resource allocation and a more productive and possibly smaller public sector. It has been argued that locally determined policies are better able to take into account the regional and local conditions. This is particularly important when decisions regarding the provision of local public goods such as infrastructure, education and health care are made (Oates 1993). Martinez-Vazquez and McNab (2003) have argued that fiscal decentralisation can provide incentives for local governments to be innovative. Brennan and Buchanan (1980) have suggested that fiscal decentralisation can lead to competition among different levels of governments which is likely to promote lower tax rates and efficient provision of public goods under revenue constraints. Tanzi (1995) and Ter-Minassian (1997) have argued that fiscal decentralisation can facilitate the development of appropriate macroeconomic policies in general and it can also help in implementation of stabilization policies in particular.

Recent studies such as Martinez-Vazquez and Rider (2005), Oates (2007) and Vo (2008a, 2008b, 2009) not only provide an excellent review of the literature but also highlight some emerging issues. For example, Martinez-Vazquez and Rider have acknowledged that the effect of fiscal decentralisation on economic growth is a complex issue which requires careful analysis. They have also argued that fiscal decentralisation can contribute

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to increased macroeconomic instability and corruption but the empirical evidence in this area is inconclusive. Oates (2007) reinforces the view that fiscal decentralisation does not always lead to better allocation of resources within the public sector. While highlighting the importance of fiscal decentralisation and its role in some developing countries, Vo has suggested a new measure of the degree of fiscal decentralisation that takes into account fiscal importance as well as fiscal autonomy of subnational governments.

This paper focuses on the link between fiscal decentralisation and economic growth. On the basis of the existing literature, it can be argued that fiscal decentralisation can affect economic growth through (1) reducing the size of government, (2) improved allocation of resources within the public sector and (3) increased competition among subnational governments. Martinez-Vazquez and Rider (2005) have argued that the lack of fiscal discipline and poor service delivery in China and India has the potential to adversely affect the sustainability of GDP growth rate of both countries. Fiscal decentralisation can take the form of expenditure as well as revenue decentralisation. Both are expected to have a positive effect on economic growth. Expenditure decentralisation contributes to economic growth by ensuring efficient allocation of public spending. Revenue decentralisation can contribute to economic growth by increasing the revenue collection, which can serve to reduce the government debt and hence its reliance of debt financing. In addition, an increase in government revenue reduces the incentive to increase the tax rates.

A number of studies have considered the link between fiscal decentralisation and economic growth. These studies have produced mixed results – some studies have found the effect of fiscal decentralisation on economic growth to be positive whereas others found no or inverse relationship between the two variables. For example, by making use of panel data involving 51 developed and developing countries for the period 1997–2001, Iimi (2005) found a positive relationship between fiscal decentralisation and GDP growth. Akai and Sakata (2002) found a positive relationship between fiscal decentralisation and GDP growth in the USA for the period 1992–1996. Lin and Liu (2000) found the impact of 1980s fiscal decentralisation on Chinese GDP growth to be positive. Zhang and Zou (1998) found the impact of fiscal decentralisation on economic growth in China to be negative (this study is based on data for the period 1980–1992). By making use of a panel dataset that involves 46 developed and developing countries for the period 1970–1989, Davoodi and Zou (1998) found a negative relationship between fiscal decentralisation and economic growth for developing countries and no relationship between fiscal decentralisation and economic growth for developed countries. Woller and Phillips (1998) found no evidence of a statistically significant relationship between GDP growth and fiscal decentralisation in 23 developed and developing countries for the period 1974–1991. Thornton (2007) found the impact of fiscal decentralisation on economic growth to be statistically insignificant for 19 OECD countries for the period 1980–2000. Carrion-i-Silvestre *et al.* (2008) have considered fiscal decentralisation in Spain. They found the impact of revenue decentralisation on economic growth to be positive, whereas the impact of expenditure decentralisation on economic growth was negative.²

While the existing studies provide ample evidence of a link between fiscal decentralisation and economic growth in both developed and developing countries, none of the existing studies have empirically examined the impact of Vietnam's fiscal decentralisation on its economic growth. By making use of a recently released panel dataset that covers 61 Vietnamese provinces, this paper attempts to empirically examine the link between fiscal decentralisation and economic growth. We examine the link between fiscal decentralisation and economic growth in Vietnam during two sample periods: (1) from 1997 to 2001, which corresponds to the introduction 1996 State Budget Law, and (2) from 2002–2007, which

corresponds to the introduction of 2002 State Budget Law. The 1996 State Budget Law led to significant fiscal decentralisation in Vietnam, whereas the 2002 State Budget Law led to further expansion of fiscal decentralisation in Vietnam.

The rest of this paper is organised as follows. Section 2 includes a brief discussion of fiscal decentralisation in Vietnam. Methodology is presented in Section 3. The empirical results are presented in Section 4, and Section 5 contains some concluding remarks.

2. Economic growth and fiscal decentralisation in Vietnam

At the end of the War in 1975, Vietnam effectively had two economic systems – a centrally planned system in the north and a market-based system in the south. However, soon after national reunification in 1975 to 1989, Vietnam remained a centrally planned economy. The introduction of the reform policy known as *Doi Moi* in 1986 marked the beginning of the opening up of the Vietnamese economy, which resulted in a significant inflow of foreign investment.³ Rapid foreign investment in Vietnam has contributed to impressive economic growth. The reform process has resulted in a general increase in the standard of living in Vietnam as measured by real GDP per capita. The GDP in Vietnam increased approximately US\$ 6.5 billion in 1990 to about US\$ 71.0 billion in 2007. In 2008, the economic growth rate was 6.18% (GSO 2009). Figure 1 shows an interesting transition in Vietnamese economy; from 2002 onwards, the contribution of the manufacturing sector to GDP is consistently larger than that of the agricultural sector.

Vietnam has four levels of government: central, provincial, district and communal. During the *Doi Moi* period, in order to facilitate efficient allocation of resources and in an attempt to meet the needs of local population, the Vietnamese government introduced an extensive fiscal decentralisation programme.⁴ The foundation of this programme was the approval of the 1996 State Budget Law and its revision in 1998. As a result, the share of subnational governments in total expenditures increased from 26% in 1992 to 36% in 1997 and to 43% in 2001 (MOF 2008). In 2001, substantial changes to the existing arrangements were made that led to further decentralisation. A new State Budget Law was approved in 2002. It has been suggested that fiscal decentralisation in Vietnam has improved transparency and brought the decision-making process closer to beneficiaries. Fiscal decentralisation in Vietnam has strengthened the public project selection process but at the same time it has considerably weakened the project appraisal and implementation

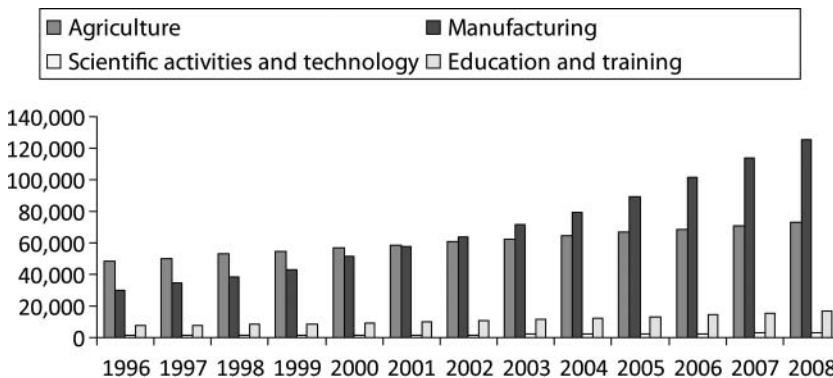


Figure 1. GDP by type of economic activity at 1994 constant prices (billion Vietnamese Dong, VND). Source: GSO (2009).

process (Vietnam Development Report 2009). The recently released Vietnam Development Report also highlights the absence of cost–benefit analysis and the lack of monitoring of project execution in some cases. The report highlights the need for a law of public investment.

The tax base and the tax rates in Vietnam are determined by the central government, which also administers the taxation policy. In addition, the central government has a big say in spending decisions taken by each of the subnational governments and hence the level of autonomy is severely restricted. Vietnam can be described as a moderately fiscally decentralised economy (Vo 2009). Vo also highlights the need for a greater autonomy at subnational level given that the existing level of autonomy in Vietnam is well below the autonomy in China, Indonesia, the Philippines and Thailand. It can however be argued that as far as developing countries are concerned, less autonomy is not necessarily a bad thing given that it can contribute to increased corruption, which remains a major concern. In other words, it can perhaps be argued that in the short term, less autonomy is not a bad policy as long as it serves to reduce corruption.

3. Methodology and data

For the purposes of the present study, we make use of a recently released panel dataset which provides annual data on 61 provinces of Vietnam for the period 1997–2007. Unfortunately, comparable provincial data on relevant variables are not available for previous years, and hence we have no choice but to restrict our empirical study to 1997–2007. In addition, the choice of the determinants of economic growth used in the empirical model specified below is also dictated by the availability of data. The panel data utilised for the purposes of this study are divided into two subsample periods: 1997–2001 and 2002–2007. Most of the data are collected from the General Statistics Office (GSO), the Ministry of Planning and Investment (MPI), the Ministry of Labour Invalids and Social Affairs (MOLISA), the Ministry of Finance (MOF) and the Ministry of Industry (MOI). See Table 1 for variable definition and data sources. All nominal variables are measured in 1994 constant prices.

Most existing empirical studies have measured the level of fiscal decentralisation as (1) the share of subnational government expenditure as a percentage of the national total and (2) the share of subnational government revenue as a percentage of the national total. The former measure can be described as expenditure decentralisation whereas the latter as revenue decentralisation. On the basis of the existing literature and given the data availability constraints, the link between economic growth and fiscal decentralisation in Vietnam is empirically examined by making use of the regression equation as follows, where G_{it} is the GDP growth rate of province i in period t and so on, and α values are the unknown population coefficients:

$$G_{it} = \alpha_0 + \alpha_1 FD_{it} + \alpha_2 E_{it} + \alpha_3 HC_{it} + \alpha_4 GAP_{it} + \alpha_5 LD_{it} + \alpha_6 L_{it} + \alpha_7 IF_{it} + \alpha_8 M_{it} + \varepsilon_{it}. \quad (1)$$

The theoretical foundation of Equation (1) is similar to that presented by Davoodi and Zou (1998) and Xie *et al.* (1999). The expected signs of the estimated coefficients are described in Table 2. As indicated earlier, the impact of fiscal decentralisation on economic growth can be negative (see Oates 2007). It is well known that the impact of inflation on the rate of economic growth can be either positive or negative. Fischer (1993), among others, has argued that macroeconomic factors affect economic growth through

Table 1. Variable definition and data sources.

Abbreviations	Variable definition	Source
G	Provincial economic growth rate (annual %).	GSO
FD	Fiscal decentralisation: two measures are used to gauge the degree of fiscal decentralisation: (1) provincial budgetary expenditure as a percentage of total budgetary expenditure and (2) provincial budgetary revenue as a percentage of total budgetary revenue. ⁵	MOF
E	Exports of goods and services per capita (expressed in thousands of VND at constant prices).	GSO
HC	Human capital: HC is proxied by the number of university and college students per thousand persons.	GSO
LD	Learning by doing (the annual manufacturing value added as a percentage of GDP is used as a proxy). ⁶	MOI
L	The growth rate of labour (annual %).	MOLISA
GAP	Technology gap: technology gap is measured as the percentage difference between the average growth of Ho Chi Minh City and other provinces. ⁷	GSO
IF	The rate of inflation (annual %).	GSO
M	Financial development is measured by the growth rate of money supply (M2).	GSO
ε_{it}	Error term.	

uncertainty. For example, uncertainty associated with high inflation can reduce the growth rate of productivity and hence its effect on economic growth can be negative. However, the traditional growth theories tend to emphasize the positive impact of inflation on capital accumulation which arises as a result of the portfolio shift. A rising inflation rate reduces the rate of return on money, which forces investors to invest in real assets that are likely to be used for further production and, hence, the impact of rising inflation on economic growth can be positive.

The empirical model specified above is estimated by making use of panel regression technique where (1) fixed effects have been taken into account and (2) correction has been made for panel heteroskedasticity and serial correlation. It is well known that panel data can better capture the dynamics of the relationship between economic variables (see Baltagi 2008).

Table 2. The expected signs of the estimated coefficients.

Independent variables	Dependent variable: provincial economic growth rate (G)
FD	+/- (see Oates 2007)
E	+
HC	+
GAP	-
LD	+
L	+
IF	+/-
M	+
	(see Fischer 1993)

4. Empirical results

Table 3 shows the estimated results for the period 1997–2001. This period corresponds to the implementation of the 1996 State Budget Law, which led to significant fiscal decentralisation in Vietnam.

The estimated results presented in Table 3 indicate that economic growth is negatively associated with expenditure decentralisation and positively associated with revenue decentralisation. The estimated coefficient of expenditure decentralisation is negative and statistically significant at 1% level, while the coefficient of revenue decentralisation is positive and statistically significant at 5% level. The estimated results support the view that revenue decentralisation has promoted economic growth in Vietnam but the negative association between expenditure decentralisation and economic growth contradicts the conventional wisdom of fiscal decentralisation in Vietnam. It is interesting to note that this pattern is consistent with the empirical work of Jin and Zou (2005) and Zhang and Zou (1998). They have argued that in the early stages of economic development, the central government may be in a better position to undertake public investment with nationwide

Table 3. The impact of fiscal decentralisation on economic growth (1997–2001).

Independent variables (1)	Dependent variable: provincial economic growth	
	Fixed effect estimation without Asian financial crisis dummy variable (2)	Fixed effect estimation with Asian financial crisis dummy variable (3)
Expenditure decentralisation (FD)	−0.602431 (−2.76)***	−0.620307 (−3.03)***
Revenue decentralisation (FD)	0.116392 (2.02)**	0.122181 (2.21)**
Exports (E)	0.000031 (0.31)	0.000026 (0.25)
Human capital (HC)	0.219293 (1.44)	0.035100 (0.19)
Technology gap (GAP)	−1.52758 (−3.21)***	−1.612200 (−10.19)***
Learning by doing (LD)	0.037520 (2.54)***	0.034302 (2.93)***
Labour growth (L)	0.599493 (2.53)***	0.645131 (3.46)***
Inflation (IF)	−0.013161 (−1.86)*	−0.020871 (−1.42)
Financial development (M)	−1.431511 (−0.55)	−1.949739 (−0.63)
Asian financial crisis (1997)		−1.514787 (−4.13)***
Constant	18.706020 (6.34)***	10.196810 (6.16)***
Number of observations	288	288
Number of groups	61	61
R ² within	0.2646	0.3128
R ² between	0.5036	0.5236
R ² overall	0.3779	0.4101

Notes: (1) Robust *t*-statistics are given in parentheses. (2) *Significant at 10%, **significant at 5% and ***significant at 1%.

externalities. The same explanation can also be applied to the case of Vietnam. Also, it has been argued that expenditure decentralisation can contribute to increased corruption, which can also explain the negative impact on economic growth.

Other important determinants of economic growth considered include exports, human capital, technology gap, learning by doing, the level of financial market development and the growth of labour force and the inflation rate. The estimated coefficient of technology gap is statistically significant at 1% level and its sign is consistent with expectations. Similarly, the coefficient of learning by doing is positive and significant at 1% level. The significant effect of learning by doing on economic growth perhaps reflects the fact that in the case of Vietnam, some production activities involve assembling only. The Vietnamese labour force appears to have benefited from knowledge spillovers through labour turnover, which has contributed to improvement in productivity thereby stimulating economic growth. The estimated coefficient of inflation rate has a negative sign and it is statistically significant at 10% level. The impact of other variables such as exports, human capital and the level of financial development on economic growth is statistically insignificant.

The Asian financial crisis of 1997 significantly affected Vietnam and hence the empirical model was re-estimated by introducing a dummy variable on the right-hand side of Equation (1). The estimated coefficient of the dummy variable is expected to be negative. Column 3 of Table 3 shows the estimated results when Equation (1) is re-estimated after inclusion of a financial crisis dummy variable. The financial crisis dummy variable has a negative sign and it is statistically significant at 1% level, which confirms that Asian financial crisis did adversely affect the economic growth of Vietnam. A careful examination of the estimated result shows that the introduction of the dummy variable has resulted in a minor change in the magnitude of the estimated coefficients without affecting their significance level.

As indicated earlier, fiscal decentralisation programme in Vietnam gained further momentum with the introduction of the 2002 State Budget Law and hence there is a need for re-evaluation of the impact of fiscal decentralisation on economic growth from 2002 onwards. In addition, the recent literature on fiscal decentralisation suggests that development investment expenditures, recurrent expenditures and intergovernmental transfers can also significantly affect economic growth in developing countries (see Jin and Zou 2005 and Vo 2006). Thus, we investigate the effects of both development investment expenditures and recurrent expenditures on economic growth as well as the effect of fiscal decentralisation on economic growth in Vietnam via the intergovernmental transfers. It is perhaps worth mentioning that the dataset utilised for the present study does not include information on development investment expenditure, recurrent expenditure and intergovernmental transfer for the period 1997–2001. As a result, we have no choice but to estimate the effect of these variables on economic growth in Vietnam only for the period 2002–2007. Both development investment and recurrent expenditures are measured as shares in total budgetary expenditures, while the intergovernmental transfer is measured by central government transfer to provinces as a percentage of total provincial expenditure.

Table 4 presents the estimated results for the period 2002–2007. These results are based on re-estimation of Equation (1). Equation (1) was re-estimated after introducing additional explanatory variables. The results reported in Table 4 have taken into account the fixed effects and correction has been made for panel heteroskedasticity and serial correlation.⁸

Column 2 of Table 4 shows that economic growth of Vietnam is negatively associated with expenditure decentralisation. The estimated coefficient of expenditure decentralisation is statistically significant at 1% level, which suggests that decentralisation of expenditure is not a good strategy and in fact the country will be better off with centralisation of

Table 4. The impact of fiscal decentralisation on economic growth (2002–2007).

Independent variables	Dependent variable: provincial economic growth		
	(2)	(3)	(4)
Expenditure decentralisation (FD)	−0.404835 (−2.97)***	−0.506447 (−4.02)***	
Expenditure decentralisation (FD) (development investment expenditures)			0.277779 (1.68)*
Expenditure decentralisation (FD) (recurrent expenditures)			−0.707242 (−3.75)*
Revenue decentralisation (FD)	0.103103 (2.83)***	0.139549 (2.66)***	0.046906 (0.68)
Intergovernmental transfer (IT)		−0.923975 (−2.15)**	
Exports (E)	0.000081 (3.54)***	0.000050 (1.81)*	0.000039 (1.62)*
Human capital (HC)	0.009790 (2.32)**	0.008576 (1.38)	0.010892 (1.86)*
Technology gap (GAP)	−1.792744 (−2.08)**	−1.287699 (−2.24)**	−1.407681 (−2.05)**
Learning by doing (LD)	0.017183 (2.49)***	0.018389 (2.74)***	0.017167 (3.06)***
Labour growth (L)	0.055401 (1.79)*	0.013165 (0.83)	0.014464 (0.96)
Inflation (IF)	0.128364 (1.99)**	0.028374 (0.92)	0.024817 (0.80)
Financial development (M)	0.941101 (1.89)*	1.518655 (4.38)***	0.300666 (1.38)
Constant	−4.460684 (−0.66)	6.55434 (2.12)**	6.493303 (2.09)**
Number of observations	322	322	322
Number of groups	60	60	60
R ² within	0.3621	0.2713	0.2458
R ² between	0.4473	0.4754	0.4648
R ² overall	0.3511	0.3563	0.3495

Notes: (1) Robust *t*-statistics are given in parentheses. (2) *Significant at 10%, **significant at 5% and ***significant at 1%.

expenditures. In other words, the estimated results appear to suggest a rolling back of expenditure decentralisation in Vietnam. The estimated coefficient of the revenue decentralisation is positive and statistically significant at 1% level. This suggests that subnational governments in Vietnam are doing a very good job at revenue collection from public. In other words, revenue collection at a local level is in fact contributing to economic growth. It can be argued that local-based revenue collectors are able to take advantage of their superior knowledge concerning the local affairs, which contributes to a healthier fiscal position and has a positive impact on economic growth. Other independent variables included are statistically significant and their sign is consistent with expectations.

It is interesting to note that impact of exports, human capital and the level of financial development on economic growth during the sample period of 1997–2001 was statistically insignificant. However, the effect of these variables on economic growth during the sample period of 2002–2007 is statistically significant. The presence of a significant positive relationship between exports and economic growth can be attributed to the export surge

following the signing of a bilateral trade agreement with the USA in July 2001. The contribution of exports to economic growth in Vietnam is likely to continue to increase given that Vietnam has joined WTO in January 2007. An increase in the productivity of human capital overtime could be attributed to the Vietnamese economy's greater integration with the international economy and increased spending on education and training. It is also possible to argue that overtime the level of financial development in Vietnam has improved and hence its effect on economic growth has become significant.

Column 3 of Table 4 indicates that the estimated coefficient of the intergovernmental transfer is negative and significant at 5% level. It can be argued that the difference between the expected revenue and minimum expenditure needed by the subnational governments, which is covered by transfers from the central government to the subnational governments, has not promoted economic growth in Vietnam. Column 3 of the table shows that the impact of exports, learning by doing and financial development on economic growth during the period 2002–2007 is positive. However, the impact of human capital, labour growth and inflation is statistically insignificant.

Column 4 of Table 4 shows that the estimated coefficient of development investment expenditures is positive and statistically significant at 10% level. In other words, development investment expenditures have contributed to economic development in Vietnam. This suggests that at the provincial level, increased investment in infrastructure, advanced technologies, education, research and development and health is making a positive contribution to economic growth. However, economic growth is negatively associated with recurrent expenditures. The estimated results suggest that at the provincial level spending on administration, price support and economic services is relatively inefficient.

5. Concluding remarks

The relationship between fiscal decentralisation and economic growth remains an issue of intense debate among policy makers and researchers. While this debate has provided rich insights into the relationship between fiscal decentralisation and economic growth, none of the existing studies have empirically considered the case of Vietnam. By making use of a recently released panel dataset, which covers 61 provinces of Vietnam, this paper considers the impact of fiscal decentralisation on economic growth in Vietnam. We consider two sample periods: 1997 to 2001 and 2002 to 2007. It is widely acknowledged that significant fiscal decentralisation took place in Vietnam from the mid-1990s and the process of fiscal decentralisation received a further boost in 2002. Following the existing literature, we consider both revenue and expenditure decentralisation. The expenditure decentralisation contributes to economic growth through increased efficiency, whereas revenue decentralisation can serve to reduce government debt and hence its reliance on debt financing.

The empirical analysis presented in this paper is based on panel regression, where fixed effects have been taken into account and corrections have been made for panel heteroskedasticity and serial correlation. The estimated results for both sample periods suggest that the effect of expenditure decentralisation on economic growth in Vietnam has been negative, whereas the effect of revenue decentralisation on economic growth has been positive. The first sample period of 1997 to 2001 corresponds to the introduction of the 1996 State Budget Law, which was revised in 1998. On the basis of the 1996 Budget Law and its revision, tax collection was localised and the provinces, districts and communes collected taxes on behalf of the central government. The decentralisation of the revenue was given a higher priority. However, as far as the spending decisions are concerned, the central government has granted very little autonomy to the subnational governments. It

can be argued that through constant interference, the central government is attempting to reduce the possibility of misuse of expenditure at subnational government level. However, this policy can also be interpreted as a lack of confidence in subnational governments. While this strategy may have helped to reduce corruption, the effect of expenditure decentralisation on economic growth in Vietnam is negative. The results presented in this paper support the view that provincial governments are efficient in collecting revenue via taxes, while the central government appears to be more efficient in spending it (Zhang and Zou 1998, Jin and Zou 2005). The negative effect of expenditure decentralisation on economic growth appears to suggest a wasteful use of public revenue. This could perhaps be, to some degree, attributed to the lack of autonomy granted by the central government.

Our empirical results based on the data for the period 2002 to 2007 suggest that revenue decentralisation has made a positive contribution to economic growth in Vietnam. As far as the expenditure decentralisation is concerned, we found that the development investment expenditures were positively associated with economic growth, which suggests that development investment expenditure decentralisation has promoted economic growth. This could perhaps be attributed to the fact that as far as investment in infrastructure, health care and education is concerned, the provincial governments have relatively more autonomy. In other words, the provincial governments have been able to efficiently utilise the development investment funds. However, the same conclusion does not apply to the recurrent expenditures. It appears that the recurrent expenditures have not contributed to economic growth in Vietnam. This can be attributed to the fact that the provincial governments in Vietnam tend to hire excessive administrative staff some of whom are poorly qualified. In addition, the provincial governments spend a significant amount of money on price support programmes for goods produced by the state-owned enterprises.

On the basis of the results presented in this paper and by drawing on the existing literature, it can be argued that in the early stages of economic development, the central government is in a better position to invest in projects of national significance because such projects tend to generate nationwide externalities. The central governments are better equipped to maintain macroeconomic stability in both the short and the long run. However, Vietnam will greatly benefit from greater expenditure decentralisation. Expenditure decentralisation in Vietnam can make a positive contribution to economic growth if steps are taken by the central government to enhance the administrative capacity of the subnational governments. This may entail developing and implementing programmes to upgrade the level of technical and administrative capacities at the district and commune levels. Such programmes are likely to improve the expenditure management skills of the subnational governments.

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Notes

1. The importance of fiscal decentralisation has also been highlighted by a number of international organizations including the Asian Development Bank (see Usui 2007).
2. The literature on fiscal decentralisation is continuously growing and the impact of fiscal decentralization on a number of variables is being considered. For example, Neyapti (2004) examines the relationship between revenue decentralisation and central bank independence. Neyapti suggests

that a negative relationship exists between revenue decentralisation and inflation when central bank is independent and local accountability exists.

3. Vietnam started attracting significant FDI from 1988, which can be attributed to promulgation of a liberal foreign investment law in 1987. FDI inflow into Vietnam increased from US\$ 0.32 billion in 1991 to approximately US\$ 8.0 billion in 2007 (GSO 2009).
4. For an excellent review of the fiscal arrangements in Vietnam from 1976 to present, see Vo (2009).
5. Similar measures of fiscal decentralisation have been used by existing studies; see, for example, Davoodi and Zou (1998), Woller and Phillips (1998), Zhang and Zou (1998), Yilmaz (1999), Lin and Liu (2000), Akai and Sakata (2002) and Thieben (2003). Vo (2009) has suggested an alternative measure which takes into account the fiscal autonomy and fiscal importance of subnational governments. The proposed measure involves geometric mean of a subnational government's share of total national revenue and total national expenditure.
6. While examining the role of FDI in enhancing economic growth in ASEAN-5 economies, Bende-Nabende *et al.* (2001) have also used manufacturing sector value added as a percentage of GDP as a proxy for learning by doing.
7. Ho Chi Minh City was recorded to have the highest economic growth in the period 1997 to 2007.
8. A useful discussion of the role of the fixed effects and correction for panel heteroskedasticity and serial correlation can be found in Baltagi (2008).

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